J_!O_U IS COMPACT UNDER FINITE COHOMOLOGICAL DIMENSION AND SUM CONTINUITY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites the lemmas recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- Sections of j_!. Sheaves, Lemma sheaves-lemma-j-shriek-modules. Hom(j_!O_U,F) = F(U).
- K-injective embedding. Injectives, Theorem injectives-theorem-K-injective-embedding-grothendieck. K-injective representatives with injective terms.
- Computing RF on acyclics. Derived Categories, Lemma derived-lemma-unbounded-right-derived. RGamma via any complex of acyclics.
- Derived products and sums. Injectives, Lemma injectives-lemma-derived-products. oplus I_î^* represents oplus K_i.

Source reference. Original reference 00A7: sources/stacks-chow/tags.txt, source lines 379–379. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 079P: sources/stacks-chow/tags.txt, source lines 9279–9279. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 07K7: sources/stacks-chow/tags.txt, source lines 9647–9647. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 07D9: sources/stacks-chow/tags.txt, source lines 9404–9404. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let X be a ringed space. Let $j : U \to X$ be the inclusion of an open. The $\mathcal{O}_X$-module $j_!\mathcal{O}_U$ is a compact object of $D(\mathcal{O}_X)$ if there exists an integer d such that*

(1) *$H^p(U, \mathcal{F}) = 0$ for all $p > d$, and*
(2) *the functors $\mathcal{F} \mapsto H^p(U, \mathcal{F})$ commute with direct sums.*

Proof. Assume (1) and (2). Since $\mathrm{Hom}(j_!\mathcal{O}_U, \mathcal{F}) = \mathcal{F}(U)$ by Sheaves, Lemma 00A7 we have $\mathrm{Hom}(j_!\mathcal{O}_U, K) = R\Gamma(U, K)$ for K in $D(\mathcal{O}_X)$. Thus we have to show that $R\Gamma(U, -)$ commutes with direct sums. The first assumption means that the functor $F = H^0(U, -)$ has finite cohomological dimension. Moreover, the second assumption implies any direct sum of injective modules is acyclic for F. Let K_i be a family of objects of $D(\mathcal{O}_X)$. Choose K-injective representatives $I_i^\bullet$ with injective terms representing K_i, see Injectives, Theorem 079P. Since we may compute RF

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by applying F to any complex of acyclics (Derived Categories, Lemma 07K7) and since $\bigoplus K_i$ is represented by $\bigoplus I_i^\bullet$ (Injectives, Lemma 07D9) we conclude that $R\Gamma(U, \bigoplus K_i)$ is represented by $\bigoplus H^0(U, I_i^\bullet)$. Hence $R\Gamma(U, -)$ commutes with direct sums as desired. $\square$